```
var macKey = new SecretKeySpec(Arrays.copyOfRange(psk, 0,
16),    #A
        "AES");    #A
var encKey = new SecretKeySpec(Arrays.copyOfRange(psk, 16,
32),    #A
        "AES");    #A

var randomIv = new byte[16];    #B
new SecureRandom().nextBytes(randomIv);    #B
var header = "Test header".getBytes();
var body = CBORObject.NewMap()
        .Add("sensor", "F5671434")
        .Add("reading", 1234).EncodeToBytes();

var siv = new SivMode();
var ciphertext = siv.encrypt(encKey, macKey, body,    #C
        header, randomIv);    #C
var plaintext = siv.decrypt(encKey, macKey, ciphertext,    #D
        header, randomIv);    #D
```

#A Load the key and split into separate MAC and encryption keys
#B Generate a random IV with the best entropy you have available
#C Encrypt the body passing the header and random IV as associated data
#D Decrypt by passing the same associated data blocks

12.4 Key distribution and management

In a normal API architecture, the problem of how keys are distributed to clients and servers is solved using a *public key infrastructure* (PKI), as you've learned in chapter 10. To recap: